HEXIS INTERIOR · GUIDE

# User guide

Interior design at true dimensions: floor plan, 3D model, furnishing, materials, windows and doors, and bill of quantities in one environment. Tap an image with ▶ to open its video.

`Plan` → `3D` → `Furnishing` → `Materials & light` → `Walkthrough` → `Cost`

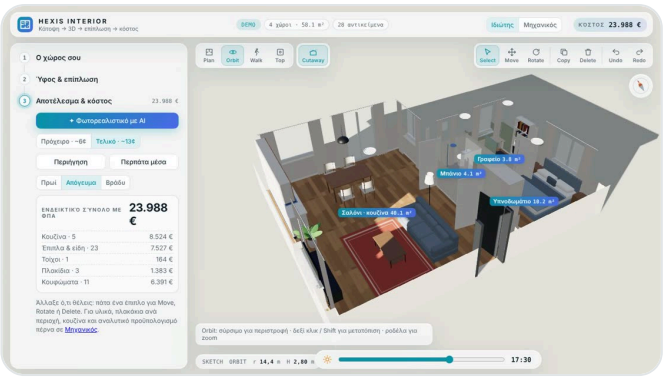
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## 1 Entry or Pro

### Three tabs at the top: 📐 Survey · ✏ Design · € Cost

- In **every** tab all tools are available: the plan toolbar (Select, Outline, Wall, Door, Window, Column, Beam, Arc, Erase, Room, Calibrate, Align, Ortho, Grid, Fit, 3D, Extras, Notes, height and wall thicknesses), Plan/Orbit/Walk, the Structure · Finishes · Drywall · Electrical · Plumbing · Furniture · Kitchen column, and the steps panel.
- 📐 **Survey** adds the survey bar: objects, photos, videos, notes, Notes, survey PDF, balcony/stair, laser.
- **€ Cost** keeps the cost table open.
- Every command lives in one place. Whatever you select in the plan shows its actions in one bar at the bottom (Properties, Delete, Undo). Undo/Redo always at top right.




Entry: three steps and the cost total.

Entry or Pro: the choice is in Settings (⚙). It changes which tools you see. Projects and drawings stay the same; your choice is remembered. The object families (Structure, Plumbing, Furniture etc.) work in every workspace, including Survey, e.g. to place sanitary ware in a bathroom. The menu stays open until you close it. **Placing in plan (2D):** pick e.g. Plumbing → WC and click in the plan. The item snaps to the nearest wall with its back to the wall (R: rotate, Alt: no snap, Esc or right-click: cancel). Sockets and switches go in with a click next to a wall and slide along it. Height: in the Electrical menu pick «standard» (socket 30, switch 110), 30, 110, 120 cm or «set it myself» (asks in cm on each click). Double-click a symbol to change its height (to the axis) at any time. Ceiling lights go in with a click where they belong. Then press Orbit or 3D to check. Finishes and Drywall still open in 3D. **Bathroom:** Plumbing has wall-hung WCs with a concealed cistern, where only the flush plate shows (white, matte black, chrome), plus separate flush plates (white, chrome, matte black, stainless, gold) at 1.00 m. There is also a **floating tiled vanity top (custom)** with 0 to 3 floating shelves below. When placed it asks for width, depth, height from floor, top thickness, shelves, shelf spacing and thickness, the tile (your own variants too), whether the shelves are tiled, and

the basin (vessel, undermount, none). Double-click to change them. At the top right, the account box always shows **PRO** or **ENTRY** in bold, so you know which edition you are using.

### Entry

- 3 steps: space, style & materials, result
- Pick and retail materials, same as Engineer
- Cost total per group
- Morning, afternoon, evening in one tap

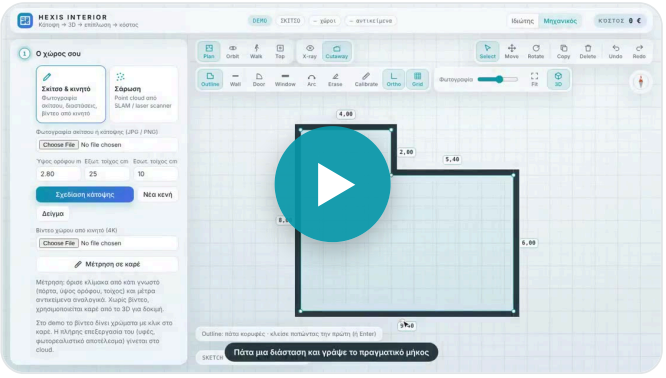
### Pro

- Scan, X-ray, Calibrate
- Materials per wall area, kitchen, windows and doors
- Detailed budget with your own prices

## 2 New project · outline

1. The app opens on an empty project, in **Outline**. To try something ready-made, press **Sample**.
2. Click the corners of the plan in order. Close the outline by clicking the first corner again or pressing `Enter`.
3. Click a dimension (e.g. `9.40`) and type the real length. The opposite side adjusts.
4. Drag a side to move it. **Undo** / **Redo** are always at the top right.
5. **Close:** tap the first vertex or the **Close outline** button in the guide at the bottom.
6. **Vertices:** with **Select** or **Outline** drag a corner; the adjacent sides stretch or shrink and stay square, and walls touching them follow. **Square corners** makes nearly square corners exact.
7. **With a finger:** tap a vertex; a panel opens at the bottom with arrows and a 1 / 5 / 10 cm step, **Coordinates** for an exact position and **90°** for a right angle.

> In step 1 you set the storey height and the thickness of outer and inner walls before going to 3D.



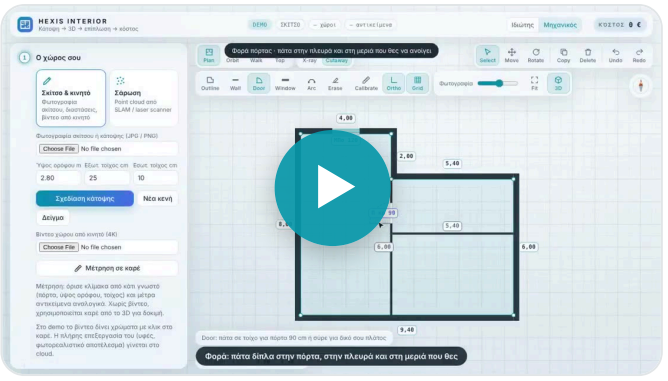

▶ Tap to play the video

An outline in five clicks, then a length correction.

## 3 Walls, windows, doors

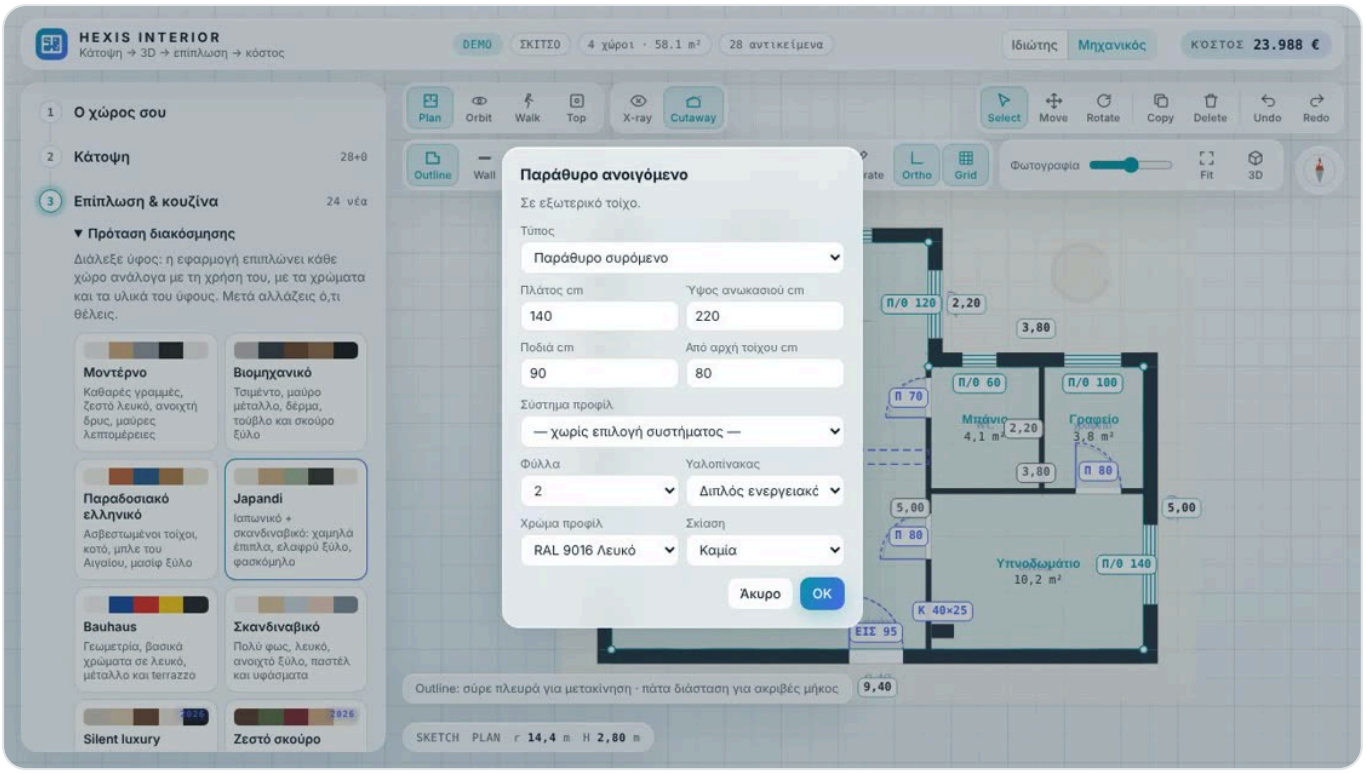
1. **Wall** : click the start and end of an inner wall. It snaps to corners and walls.
2. **Window** : click on a wall. The app asks for the type: casement, tilt-and-turn, sliding or fixed window, or a balcony door that is hinged, sliding, lift-and-slide or minimal.
3. In the same dialog you choose the profile system (ALUMIL, EUROPA, ELVIAL), sashes, glazing, RAL colour, anodised or wood-effect finish, and shading.
4. **Door** : for an interior door it asks whether it has a **perimeter frame** (frame and architraves) or **none**, with a concealed frame flush with the wall.
5. **Everything with Select :** click a door or window to select it, drag to slide it along the wall ( `Alt` for 1 mm steps), the orange distances to the wall ends take an exact position, `Delete` removes it.
6. **Swing direction:** with the door selected, click again on the side and at the end you want it to open towards. Clicking on the opening itself cycles through the four swings.
7. **Moving a wall:** with **Select** tap and drag the wall. The **clear distances** to the opposite walls on each side are shown; tap one for an exact value. In the panel at the bottom: arrows with a 1 / 5 / 10 cm step, **Distance** , **Delete** . On a computer also with the keyboard arrows.

> Click an opening's label (e.g. `BD 140` ) to change anything. **Erase** removes it.

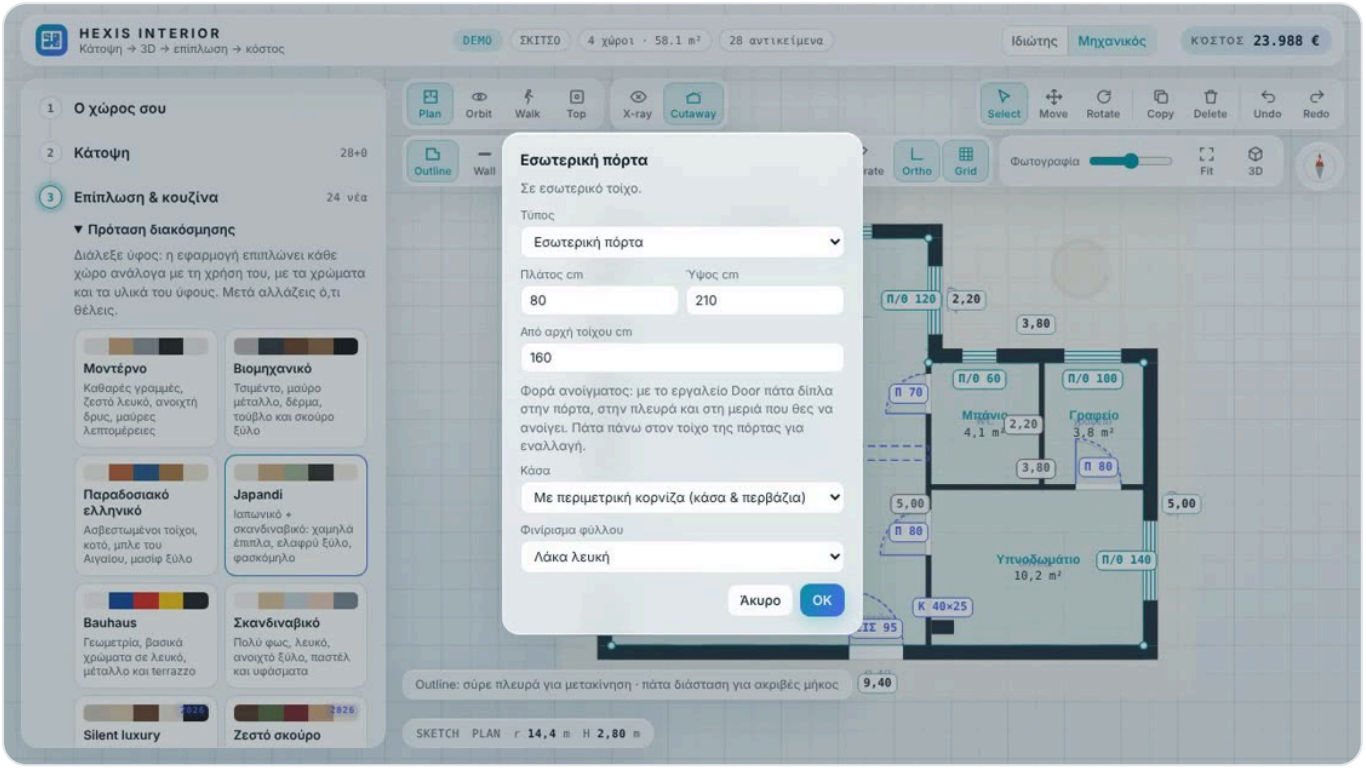



▶ **Tap to play the video**

Inner walls, a sliding balcony door, a frameless door and its swing.




Window: type, system, sashes, glazing, colour, shading.

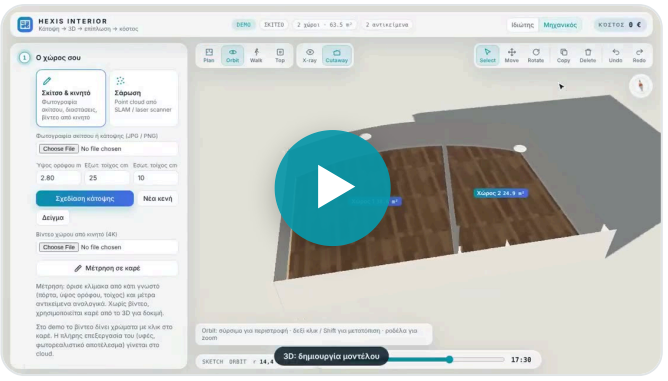



Interior door: with frame or concealed frame, and finish.

## 4 Curved & slanted walls

1. **Arc** : drag a side of the outline outwards or inwards. Click it to type the exact rise in metres.
2. Turn off **Ortho** for walls at any angle, outer or inner.
3. Press **3D** . Curved walls are built smoothly, with openings that follow the curve.

> An inner wall separates two rooms only if both its ends touch walls. On a curved wall, bring the end inside the wall's thickness.



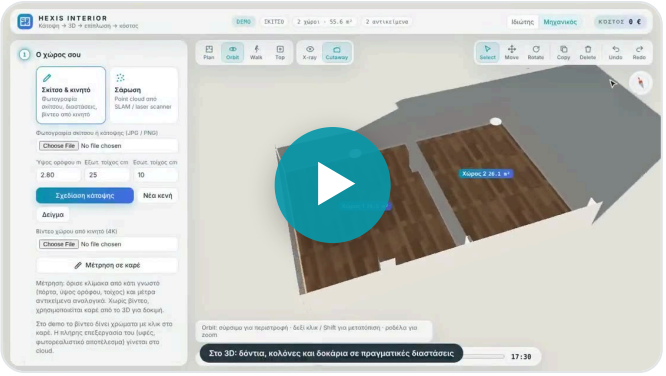

▶ **Tap to play the video**

A curved side, a slanted inner wall and the 3D.

## 5 Columns, nibs & beams

1. **Column**: click next to a wall or in free space. Pick a section type: **Rectangular**, **Shear wall**, **L-shaped**, **T-shaped** or **Circular**. Standard sections are ready (e.g. 40×40, 100×25, L 60×60/25, Ø40); you only type the dimensions and see the shape with its dimensions live. Next to a wall, the side marked “wall” sits on it and the wall stops exactly at the column. For an L, **Mirror** puts the leg on the right. Away from a wall you also give an angle. Column concrete goes into the cost.
2. **Wall build-up · ETICS** (below the wall thicknesses): pick the brick (hollow 6/9/12 holes, blocks, solid, aerated concrete 10–25, thermal bricks 20–30), single or double masonry with cavity insulation, external insulation 0–12 cm (white or graphite EPS, XPS, stone wool) and plaster. You see a section with every layer and its thickness, the total thickness and the **U-value** against the limit of the climate zone (A–D), plus the U at columns (thermal bridge). **Apply to project** sets the external and interior wall thickness automatically. The plan shows the layers hatched (when zoomed in), and the cost gets bricks, mortar, plasters, insulation boards, adhesive, anchors, mesh, primer, finish and labour. At columns the plaster wraps around continuously with the wall, with no seam, and a 30 cm glass-fibre mesh strip is added to the quantities at every concrete–masonry joint (columns and beams) so the plaster does not crack there. The U calculation is indicative; for the energy study use the manufacturer's values.
3. **Beam**: click the start and end of the beam, then its width and drop below the ceiling.
4. In plan the beam is dashed, as in drawings. In 3D it hangs from the ceiling.

> They count everywhere: room areas exclude them, nibs are painted and tiled like walls, and AI furnishing does not put tall wardrobes under a low beam.

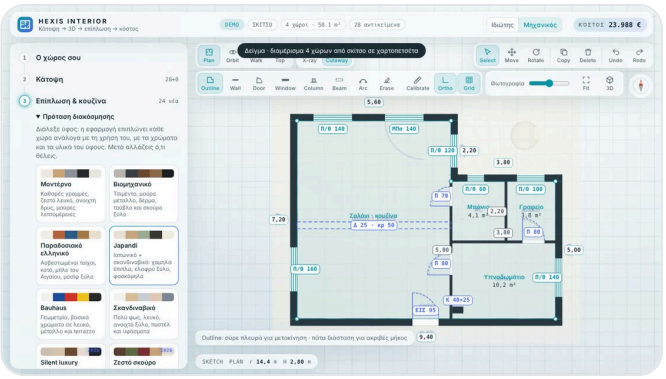



▶ Tap to play the video

A 45×30 nib on an outer wall, a column on an inner one, a beam with a 45 cm drop.

## 6 From a sketch, PDF or DXF

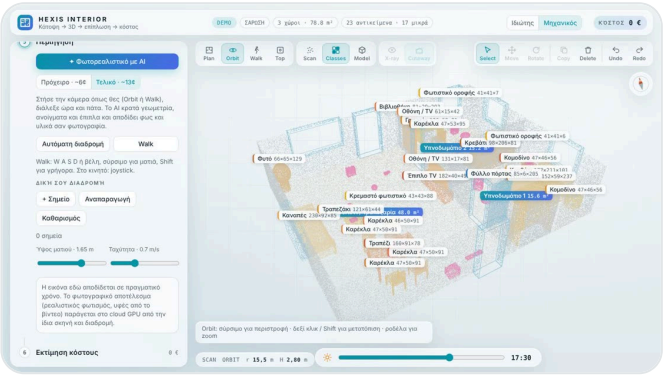
1. Step 1 → upload a photo or a **PDF** of a plan, sketch or furniture/kitchen drawing. For a multi-page PDF you choose the page.
2. **Calibrate** : click both ends of a known dimension and type its length.
3. **DXF (Pro):** upload a plan as DXF. Choose units (m, cm, mm, detected automatically) and layers, and the drawing is placed at 1:1, with no Calibrate. Click inside the inner face of the outer walls and the outline is created from the closed polyline, arcs included. If you click the outer face, enter the wall thickness and it is offset inwards. Outline, Wall, Door, Window snap to the DXF's vertices and lines.
4. If the plan already exists, e.g. you bring the cabinet maker's PDF: **Align** → click a point on the drawing, then the same point on the plan, and again for a second point. The drawing moves, rotates and scales onto your plan.
5. Tick “Also show the drawing in 3D, on the floor” and switch to **Top** : place the kitchen and furniture exactly on the drawing with Move / Rotate.
6. Draw on top with Outline, Wall, Door, Window. The `Photo` slider in the bar sets its opacity.
7. **Video per room:** once the 3D exists, step 1 lists every room with a **Video** button. Film each room separately (stand at the door, slow 360º, 20–40 s) and upload it to its room. From the video you pick colours for that room's walls and measure on frames. Four frames go into the PDF automatically as “Existing condition”; **+ Frame for the PDF** adds any frame you like.



The sample: a plan drawn over a sketch.

## 7 From a scan (Pro)

1. Step 1 → **Scan** : load a point cloud in LAS, PLY or XYZ, from SLAM or a laser scanner.
2. Step 2 → **Auto-recognition** : floor, ceiling, walls, doors, windows, rooms, furniture and small objects come out as separate entities.
3. **Scan** / **Classes** / **Model** switch the view; **X-ray** shows through the walls.

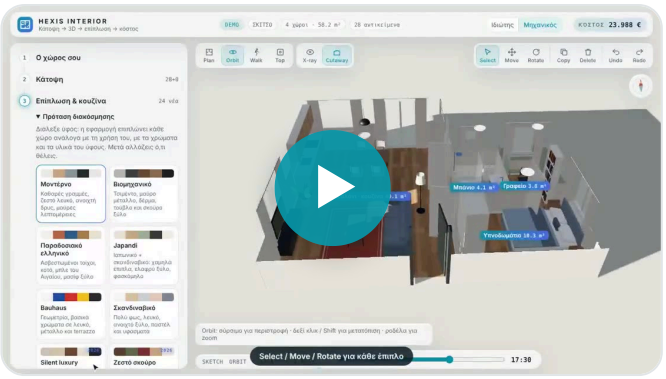


Detection from a scan, with the dimensions of each object.

## 8 AI furnishing

1. Step 3 → choose a style: Modern, Industrial, Traditional Greek, Japandi, Bauhaus, Scandinavian, and the 2026 trends from international magazines: Silent luxury, Warm dark, Neo Deco, Found luxury, Colour drenching, Coastal chic, Summer silver, Statement stone, Curves & warm minimal, At-home retreat, Refined maximalism, Book-lined, Vintage barber · Speakeasy.
2. Write what you want, e.g. “desk in the small room, table for 6”. Press **✦ Improve description** : the AI turns it into a precise professional brief (spaces, furniture, materials and colours, lighting in K and lumen, what to avoid) without losing anything you wrote. Edit it freely; **↺ Original text** restores yours. Your brief always takes priority over the style. Improving uses no units.
3. Press **✦ AI layout** . The designer sees the real geometry, the openings, columns and beams, names the rooms and places furniture with collision checks.
4. Change anything with **Select** , **Move** , **Rotate** , **Copy** , **Delete** . **Undo** reverts the whole proposal.

> Each style has its own signature pieces, layered lighting with lumen, rugs, plants and decor, and respects the space names. “Quick suggestion (rules)” does the same without AI, using rules. It is instant, but simpler in slanted rooms.



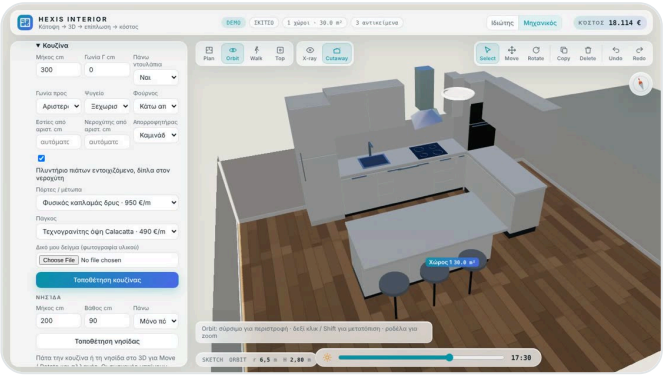

▶ Tap to play the video

Style, instructions and a furnishing proposal.

## 9 Kitchen, L-corner, island & appliances

1. Step 3 → **Kitchen** : length, **L-return cm** and **Return to** left or right for a corner kitchen.
2. Appliances: built-in fridge left or right, oven under the hob or in a tall unit, chimney or built-in hood, dishwasher.
3. Hob and sink position in cm from the left end; leave empty for automatic placement.
4. Fronts (veneer, laminate, lacquer) and worktop (engineered stone, quartz, wood), or your own sample from a photo.
5. **Island placement** : length, depth and what sits on it: worktop, hob, sink or a bar with stools.

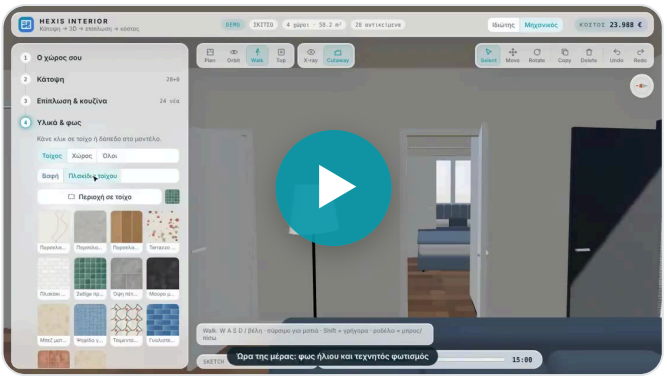
> Click the kitchen in 3D and change values in the form; it updates instantly, together with the cost.




L-return to the right, fridge on the left, oven column, chimney hood, bar island.

## 10 Materials, tiles, light

1. Press **Pick** at the top right, then a wall or floor. Pick sees through furniture and shows the surface's current material. Choose **Wall**, **Room** or **All**.
2. **Paint** from the palette or your own colour, or **Wall tiles** with a height.
3. **Your own tile variant:** in Wall tiles open «Your own variant». Pick a texture (plain, cement, stone, marble, terrazzo, zellige, wood, mosaic, brick, micro-pebble, microcement), a finish (matte, satin, gloss, honed, polished, brushed, sandblasted, bush-hammered), a colour from 24 shades or your own, the size, the grout and staggering. Terrazzo has 72 ready shades in four groups (Classics, Neutrals, Colours, Beige), plus chip palettes, density and chip size. You can also type a product/code and a catalogue link. «Create & apply» adds it to the list for walls and floors.
4. **Area on wall**: drag a rectangle (e.g. 2.00 × 1.20 m) to tile only that part.
5. **Retail material:** open “Retail material (your own product)”, enter the company and series/code, tile size, price in €/m² and upload a photo of one tile from the catalogue. For paint, the colour code and price. It joins the list under its name and counts in the cost.
6. **High-fidelity materials:** in hexis-app, at High or Ultra quality, floors (light and dark wood, marble, screed) get real photo textures with relief and roughness, and walls, fabrics and furniture wood get real relief while keeping their colour. Textures are free to use (CC0). Turn off with “High-fidelity materials”.
7. **Wall junctions:** at the floor **Skirting** (height in cm), **Shadow gap with aluminium profile** (10, 15 or 20 mm) or **None**; at the ceiling a shadow gap with aluminium profile or a plaster cornice where the wall begins. They appear in 3D, in the bill of quantities in metres and in the PDF materials per room.
8. Materials open only with **Pick**. With **Select** you can click walls and floors without the menu jumping, so no material is applied by mistake.
9. Floors per room, time and season for the sun, lighting intensity and colour temperature (default 4000 K, neutral white), image quality **Fast** / **High** / **Ultra**.



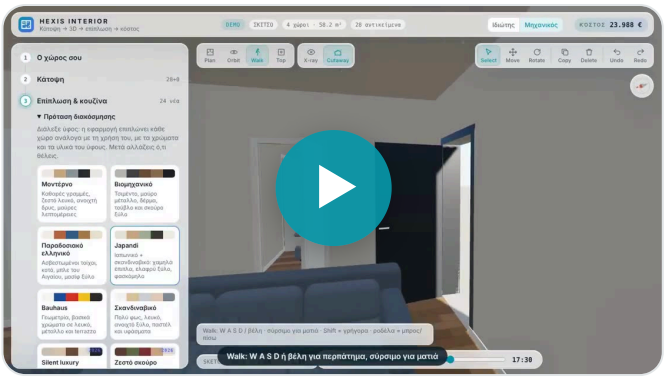

▶ Tap to play the video

Wall colour, tiles and light from noon to evening.

## 11 Walkthrough & photoreal

1. **Go into a room:** double-click inside a room in the plan, click the room label in 3D, or use the **Go into room...** list next to Walk. You stand at the door, facing the centre, and see the whole room, with a wider lens in small rooms. **Room** shows the whole room from above. At doors, Walk pulls you onto the door axis so you pass through easily.
2. **Walk** : `W` `A` `S` `D` or arrows, drag to look, `Shift` to go faster. On a phone, with the joystick.
3. Step 5 → **Auto path**, or your own with **+ Point**.
4. Set up the camera and press **✦ Photoreal with AI**. **Draft** for tests, **Final** for presentation. Choose how freely the AI works: **Faithful** (light and materials only), **Photoreal** (simple 3D furniture becomes real products with fabrics, wood, stone and details, in the same positions and sizes), **Creative** (the 3D is a layout guide and the AI styles the room like a magazine shoot). Products you added from shops ("Put it in the project") that are in view, up to 6, are sent with their photo and position, so the AI renders those actual pieces. Final is rendered at 2K, in your screen proportions. Geometry and furniture stay the same; light and materials change.
5. Compare with the 3D and press **Download**.
6. **Remodeling from a photo:** **✦ Remodeling from a photo** → upload a plain photo of one side, choose **Interior** or **Building facade**, a style (for facades: Mediterranean white, Mixed natural cladding, Dark & warm, Contemporary minimal, Biophilic, Restored traditional) and what changes; **✦ Improve description** works here too. The shot, perspective, walls and openings stay the same. No plan or 3D needed. Charged like one photoreal image.
7. **On site with AR:** choose the kitchen, the selected item, all furniture or the whole house and press **AR · View in the room**. On a phone or tablet it appears at real size in the room through the camera: tap the floor, drag to position, two fingers to rotate. On a computer you see the model in 3D.
8. **● Record video** records what the screen shows until you stop it; **Walkthrough video** records the whole walkthrough automatically.

> The AI may slightly "beautify" a piece of furniture or a texture. Dimensions and cost always come from the 3D.




▶ Tap to play the video

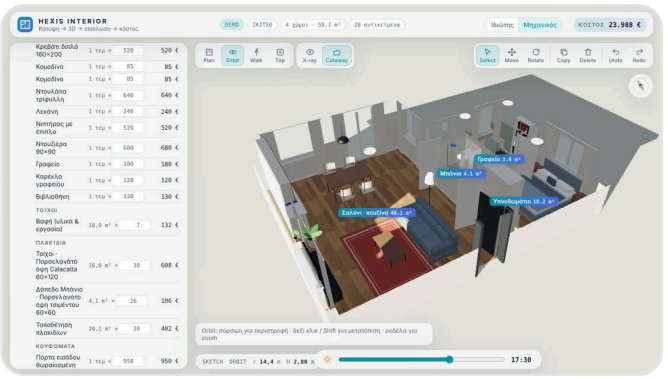
Walking through the house.

## 12 Cost

The cost at the top right updates with every change. In step 6 you see the detail: furniture, kitchen, appliances, windows and doors, paint, tiles, floors, with or without 24% VAT.

Prices are indicative. Enter your own unit price, e.g. from a supplier's quote, and the total changes instantly.

> **EL / EN** at the top right switches the language of the app and of the PDF; it opens in the browser's language automatically.



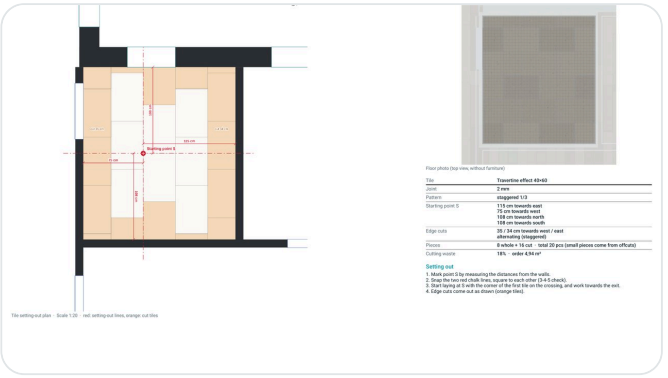

Detailed estimate with editable prices.

## 13 PDF: drawings, elevations & setting out

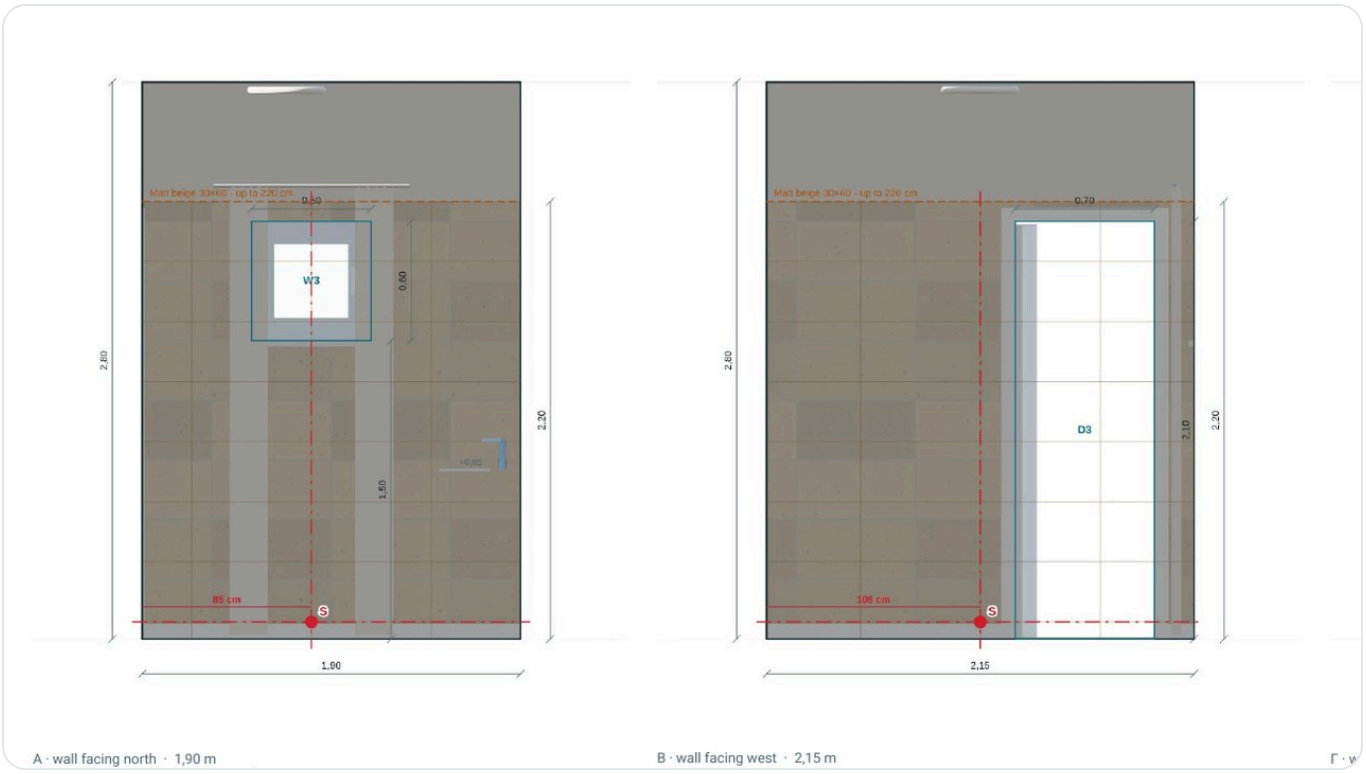
Step 6 → enter the project or client, choose which rooms get elevations (wet rooms, all or none) and press **PDF · scaled plans & bill of quantities**. Output is A3:

1. Furniture and openings plan and floor finishes plan, at 1:50 or 1:100.
2. Window and door schedule with codes, system, colour and swing.
3. **Elevations** for bathroom and kitchen (or all rooms): every wall at 1:25 or 1:50, with dimensions, openings, tile height and furniture heights. Then an **axonometric** of the room.
4. **Floor per room:** a top-view photo of the floor without furniture. Where it is tiled, a **tile setting-out plan** with the **starting point S**, the two setting-out lines, distances from the walls, edge cuts, joint, pattern, pieces and waste.
5. On tiled elevations, the wall's S: a vertical starting line (on the window axis or with symmetrical cuts) and a ledger for the first row, so the top row comes out whole.
6. “Existing condition” frames from each room's video, 4 photos per room, materials per room and the full bill of quantities with code, unit, quantity, waste, price and total.

> Setting out is chosen so that cuts on opposite sides are equal and as large as possible. Print at 100%, without “fit to page”, to measure with a scale rule.



Bathroom floor: starting point S, setting-out lines, cuts and photo.

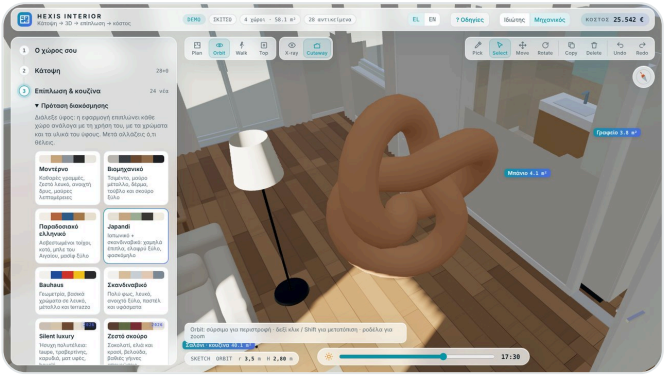



Bathroom elevations at 1:25, with the starting line and the ledger for the first row of tiles.

## 14 3D file viewer (Pro)

1. Step 1 → `3D file viewer` → choose a **GLB, glTF, FBX, OBJ (with MTL and textures), DAE, 3DS, 3DM, STL, PLY or DXF** file. Select its texture files at the same time if they are separate.
2. Units and up axis are detected automatically; if the model looks huge or lies on its side, change `Units` or `Up axis`.
3. The model goes to the centre of the plan or to its original coordinates. View it with Orbit, Top, X-ray and Walk, and send it to AR on a phone.
4. `Hide my own model` to see only the file; `Zoom to` and `Remove`.

> From a 3D program that saves in its own file type, export to GLB, FBX, OBJ or DAE first. GLB keeps materials and textures best.




A GLB file inside the house model.

## 15 Object library & collisions

1. Step 3 → **Object library**: choose a **Space** (Home, Bathroom · WC, Indoor plants, Outdoor, Decor · vases, Office, Engineering office, Clinic · dental, Shop, Barber · hair salon, Restaurant · café · bar, Lighting) or type in **Search**, e.g. “stool”.
2. Bathroom: floor-standing and wall-hung WCs, bidets, vanity basins, bathtubs, shower enclosures, walk-in, towel radiators, radiators, water heater, washing machine.
3. Press an item: it goes into a suitable space and you drag it with **Move**.
4. **Real product:** on the object's card click the title and type brand and code; change size, colour and price to match the catalogue.
5. **In shops:** every object's card has buttons for 3–4 online shops chosen by type (furniture, sanitary ware, appliances, lighting, garden, professional equipment), with a ready search by name and size. With **✦ Find in shops** the AI searches the web first and suggests, inside the app, up to 4 real products from different Greek shops, with photo, price and size; **Put it in the project** replaces the object: name, shop, price, size, colour from the photo, and the product photo stays on its card. You do not need to open the shop. It uses one AI use; if nothing reliable is found, it is refunded.
6. **Wardrobes:** a “Wardrobes” category with 1- to 6-door units (50–300 cm), with top cupboards, with mirror, sliding, built-in floor-to-ceiling, children's, chest of drawers and shoe cabinet. On the wardrobe card you change the number of doors, hinged or sliding, top cupboards and mirror.
7. **Relief / mural** (Decor): carved plaster, painted mural or patterned wallpaper, at any size on the wall; counts in the cost. New tiles: black & white checkerboard and exposed brick.
8. **Collisions:** objects do not go into each other or into walls. Chairs and stools fit under tables and counters. Hold `Alt` to move freely.
9. **Real 3D models:** with “Real 3D models” (step 3), sofas, armchairs, chairs, tables, cabinets, shelves, plants, lights, rugs etc. appear as scanned, photoreal models from a free library (CC0), at the item's dimensions. On the card: **3D model** for another variant or “Simple shape”, **↻ 90°** if the model faces the wrong way, and **My own GLB…** for the manufacturer's model. Collisions, cost and take-off stay the same.

## 16 Spaces: name and use

1. In the plan press **Room** and click inside each space.
2. Pick a name from the list (Living room, Kitchen, Bathroom, WC, Bedroom, Office, Dental surgery, Shop, Veranda…) or type your own. The name shows on the plan with its use below.
3. From the name the app knows what belongs: no bed or bookcase in a bathroom, no WC in a bedroom. If you pick something that does not fit, it goes to the right space.
4. The library opens on the category of the space you named.
5. **Ready room types:** in the room name, tap a button: Living room, Kitchen, Dining room, Bedroom, Children's room, Bathroom, WC, Hall, Corridor, Entrance, Study, Storage, Laundry, Semi-open space, Veranda, Balcony, Shop. For anything else, type it under “Other”.

> New space outside the original outline: drag the side outwards with **Outline**, divide with **Wall** and name it with **Room**.

## 17 Light fittings and light

1. Step 4 → **Light fittings**: choose a type (Recessed spot, Ceiling light, Pendant, Wall sconce, Floor lamp), **luminous flux in lumen** and light colour (2700–6000 K).
2. Press **+ Place by click** and click the floor under each fitting; spots and ceiling lights go on the ceiling. For a sconce, click the wall. `Esc` finishes.
3. Select a fitting to change its lumen and K on its card.
4. **LED strip:** choose "LED strip", enter **lm per metre** and pick a position: ceiling cove (concealed), suspended-ceiling recess, under cabinets, skirting, or shelf/niche at your own height. Clicking a wall runs the strip along its full length; with "Around the whole room" it goes on every wall of the room. It counts in the cost per metre.
5. **Remove automatic** deletes the fittings the app added.
6. **Auto exposure:** the image adjusts itself like a camera, so it does not wash out with many lights in a small room or a low sun, nor turn dark. In step 4, **Image brightness** makes it lighter or darker, and the **Auto exposure** switch turns it off.
7. **Light from video**: stop the video on a frame with the lights on; the app estimates colour temperature (K) and intensity. Positions are placed by click.

> Strip for concealed light: 600–1000 lm/m · task light under cabinets: 1000–1500 lm/m. Typical: living room 150 lm/m², bedroom 100 lm/m², bathroom 200 lm/m², office 300 lm/m².

## 18 Drywall · walls & ceilings

1. Step 4 → **Drywall · walls & ceilings**: choose the **Construction**, enter dimensions and press **+ Draw** . **Esc** ends.
2. **Wall: Wall lining**, **Lining with niche** (a recess with its own W × H), **Drywall shelf**, **Box-out**: drag a rectangle on the wall. **Thickness / projection** sets how far it comes out of the wall.
3. **Column cladding:** click near a column of the plan; it is clad all round, full height.
4. **Ceiling: Suspended ceiling (room)** = click inside a room; it drops by the **Drop from ceiling**. **Perimeter drop with concealed lighting** = click in a room; a band all round, **Band width** wide, with LED on its inner edge.
5. **Ceiling bulkhead** and **LED ceiling channel** (a light line in the suspended ceiling): drag on the floor to set position and length.
6. **Curtain / blind pocket:** click the window wall; it runs its full length, at the ceiling.
7. With **Concealed LED lighting** ticked, niches, shelves, perimeter drops and curtain pockets get an LED strip that lights the 3D.
8. Each construction is an object: select it for dimensions and price, **Move** , **Delete** , **Undo** . It counts in the cost per m² or per metre, LED strip included.
9. **Board per room:** bathroom, WC and laundry always get **moisture-resistant (green)** board; other rooms **standard (beige)**. Change it under "Drywall" in the panel or the card.
10. **Frame:** every construction has a metal frame with CW 50 studs and UW 50 tracks at **60 or 40 cm**. Suspended ceilings use CD 60/27, a UD perimeter track and hangers. With tile or marble the studs go to 40 cm automatically.
11. **View** **Frame** · **Board** · **Finish** : only the frame (board faded), the board without finish, or the final material. The builder sees what is behind the surface.
12. **Final finish** on the card: paint in any colour, a tile from the list, marble / stone or wood, at real scale.
13. **Special constructions:** bench / seating, table, desk, bookcase, TV wall unit, fireplace, cabinet. Choose it, enter the height and drag the length and depth on the floor; the back snaps to the nearest wall. Change dimensions on the card.
14. **Take-off:** on the card, "Take-off · materials & labour": m² of board, metres of profiles, hangers, corner beads, screws, compound, tape, and the cost of materials, labour and finish. The project cost lists them all together, with unit prices you can change.

> Constructions sit in front of walls and below the ceiling; they do not cut the wall. For a niche inside a wall, use **Lining with niche** with a thickness at least the depth you want.

## 19 Electrical · sockets, switches, lights

1. Left side of the 3D → **Electrical**. At the top choose the **series** (e.g. BTicino Living Now, Legrand Valena Life, Schneider Sedna), **colour** and **height**: standard (socket 30 cm, switch 110 cm), 110 cm for the kitchen worktop, or where you click.
2. Press **Socket**, **Double socket**, **Switch**, **Dimmer** etc. and click the wall, a drywall lining or the kitchen backsplash. `Esc` ends.
3. A socket or switch on tiles sits on the tile surface and a **tile cut** is added to the cost.
4. A **switch** links itself to the room's lights that have no switch yet. To change it: card → **Link to lights** and click each light (click again = unlink).
5. In **Walk**, clicking a switch turns its lights on and off.
6. Light fittings (spot, ceiling light, pendant, wall light, floor lamp, LED strip) are in the same family.
7. On the card: series, colour, height from floor, and **In shops** with an electrical supplier (Kafkas) and others; the AI searches first and brings the product into the project.
8. In the cost: every socket and switch at its series price, an **electrical point** (line, conduit, box, connection) for every point and light, and tile cuts.

## 20 Tool families · buttons

1. **They appear when you need them:** in the floor plan you only see drawing tools, with the guide **Outline › Walls › Doors & windows › Rooms › 3D & furnishing** at the bottom; the next step is coloured and its button pulses. “Not needed” skips a step. Tool families appear only once the 3D exists; on a phone they also hide in Walk.
2. Top right, the basics are always there: **Select** **Move** **Rotate** **Pan** **Copy** **Delete** **Undo** **Redo**. With **Pan**, dragging moves the view.
3. Left side of the 3D, the **families**: **Structure** (outline, wall, door, window, column, beam), **Finishes** (colours, tiles, brick, floors, plaster, shadow gap), **Drywall**, **Electrical**, **Plumbing** (WCs, wall-hung WCs, basins, mirrors, cabins, walk-in, taps, ceiling shower, bathtubs, water heater), **Furniture**, **Kitchen**.
4. Press a family; a panel opens with its tools. Press a tool and work in the 3D. The panel closes by itself after a few seconds or with `Esc`.
5. Buttons are raised: they “sink” when pressed. The active tool stays pressed.
6. The selected item's card (right) hides by itself after 5 seconds without use. It comes back with ⓘ, with the mouse at the right edge or with a new selection.
7. **On a phone:** the families are one row of icons above “Steps”. In the floor plan, Move/Rotate are hidden to save space. In landscape, views and tools share one row. Swipe the top bar sideways for language, mode and cost.

## 21 Objects from the video

1. Step 1 → **Video per room**: upload a video of the room (360°, slowly).
2. Press **✦ Objects from the video**. The AI looks at the frames, recognises furniture, sanitary ware, appliances, radiators, lamps, plants and decor, and places them in 3D at their sizes.
3. Positions come from doors and windows, so they are approximate: adjust with **Move**. A new recognition in the same room replaces the previous one; **Undo** reverts it.
4. Add frames with **+ Frame for the PDF** for a better result (up to 8).

## 22 Account, key and balance

1. **Projects:** the button with the project name, top right, opens **Projects**: ☁ Save to the cloud (also `Ctrl` + `S`), Save as..., + New project, a list of your projects with thumbnails, and a ⬇ .hexis file for a backup or a colleague. Everything is also saved automatically on the device: if the page closes, you continue where you left off. Every project, and every reopening of the app, starts in the plan; the 3D model is already built, one click on Orbit away. Cloud projects open from any device you sign in on.
2. **Client link:** in **Projects**, 🔗 Client link → + New link. Send the link by WhatsApp, Viber or email. Your client opens it on their phone with no account: they go into the rooms, use Walk, see the photos (every photoreal render you make is kept in the project, up to 8) and, if you allow it, the cost per group with and without VAT. They send you a 💬 Comment or an ✓ Approve, which appear in the list of links together with the view count. A red badge on the project button shows new comments. A link expires after 90 days; Revoke closes it at once. The client always sees the latest saved version of the project.
3. **Multiple screens:** the Screens button at the top sends the **steps** (Your space, Floor plan, Furnishing...) to a second screen and the **results** (cost by group, room areas, wall/U, details) to a third. Drawing stays alone on the main screen. Results update live. The first time, the browser asks to allow window management and pop-ups. Closing the steps window brings the steps back.
4. **Install on your computer:** the Install button at the top (in Chrome or Edge) installs HEXIS Interior as an app. It opens in its own window without an address bar and gets an icon on the desktop, taskbar and Start menu. It updates itself and keeps the same security (key, device). On iPhone/iPad: Share → «Add to Home Screen».
5. **Left panel width:** drag the right edge of the steps panel to make it wider or narrower, like property palettes in CAD. The width is kept. Double-click the edge to reset it.
6. **Floating toolbars:** the toolbars (views, Pick/Select/Move..., families, plan tools) have a ⠿ grip. Grab it and move them wherever suits you. Their position is kept, and double-clicking the grip puts them back.
7. **Esc:** cancels any active command, as in CAD: wall or outline drawing, placing a light, socket or drywall, painting, an open family. It returns to **Select** and clears the active button. If a dialog is open, Esc closes it first.
8. **Phone and tablet:** on the right of the screen there are always ↶ Undo, ↷ Redo, Delete and ✕ Esc buttons, in the plan and in 3D. To delete, tap the element first (outline vertex, wall, opening, column, object), then Delete. With nothing selected while you are still drawing an outline, Delete removes the last point.
9. 📐 **Survey with your phone (instead of a paper sketch):** the 📐 Survey tab at the top opens the survey flow.
    1. **Outline:** tap the corners or type the lengths you measure (e.g. with a laser) and tap an arrow. Finish with ✓ Close.
    2. The survey bar opens with big buttons:
        - **Wall:** start and end; snaps to the other walls.
        - **Door** and **Window:** tap the wall and type width, height, sill and distance from the corner.
        - **Column**.
        - **Object:** WC, basin, shower, bathtub, radiator, washer, water heater, bed, wardrobe, sofa, table, desk. You drag it into place.
        - **Room name**.
        - 📷 **Photos** with a note, kept inside the project.
    3. The bar shows how many rooms, openings and photos you have. ✓ See in 3D shows the result at once.
10. 📏 **Laser distance meter (Bluetooth):** the 📏 button at the top, in the Survey tab, connects a Bluetooth distance meter, e.g. Leica DISTO X5. It works on Android and on computers with Chrome or Edge; on iPhone the browser does not allow it.

- First pick your **model** from the list: Leica DISTO (X3, X4, X5, X6, D1, D2, D110, D510, D810, S910 and more) and Bosch GLM / PLR C connect directly. Hilti, Stabila, DeWalt, Milwaukee, Makita, Fluke, Mileseey, UNI-T and Huepar are a trial connection. If no reading arrives, put the meter in **keyboard mode** and choose «Any meter in keyboard mode»: the value is typed in as if from a keyboard. Your choice is remembered.
- Turn on Bluetooth on the meter and the phone, and pick the device from the list.
- Press measure on the meter. The value goes into the field you tapped (in cm where the field is in cm, e.g. door width) or, if no field is tapped, into the outline side length. Then tap an arrow.
- The button turns green while connected and shows the last reading. If the connection drops, it reconnects by itself.
- Close the manufacturer's app if it is open, because it holds the connection.

11. 📷 🎥 **Photos and videos with a position:** in survey press 📷 Photo or 🎥 Video, write a note, then tap the spot in the plan where it was taken. An icon appears in the plan; tap it to see the photo or play the video. 🗂 opens the gallery. Photos stay inside the project; videos stay on the device where they were recorded.
12. **Length with arithmetic (corners with no opposite surface):** next to the side length there are + and −. E.g. measure the full length 6.10, press −, measure the extra part 1.25 and the bar shows «= 4.85 m». Then tap the arrow. You can also type it, e.g. «3.40+0.35». The direction you hold the laser doesn't matter: the reading gives only the length and the arrow gives the direction in the plan.
13. **Closing gap:** while you type lengths, the bar shows how far the end is from the start. If it is up to 15 cm, «Close» distributes it proportionally over the sides, like a traverse, without adding an extra corner.
14. ✒ **Tablet with a pen (e.g. Samsung with S Pen):** as soon as the pen touches the screen, the app detects it.
    - **Pen** = precise drawing and selecting (corners, walls, openings, grips).
    - **Finger** = only pan (one finger) and zoom (two fingers). Your palm or finger won't add corners by mistake.
    - **Line sketching (pen only):** with the Outline or Wall tools draw lines as with a pen on paper. Each line is straightened, squared when nearly horizontal or vertical, and snaps to the ends of the others. Corners that fall short or overshoot are trimmed or extended so the lines meet. When the lines close they become the outline; after that every line inside becomes an interior wall, trimmed onto the walls. A short tap with the pen works as before (corner by tapping). ↶ Back or Undo removes the last line. Fix lengths afterwards by tapping the dimension.
    - **Pen button** + tap = Esc: cancels the running command and returns to Select.
    - ✍ **Handwriting** ( Notes on the plan toolbar or ✍ Notes on the survey bar): write or sketch on the plan in colours, thin or thick, with pressure. Eraser, Clear, Undo. Handwriting is saved in the project and appears in the overview plan of the survey report PDF.
    - In text fields (notes, room names) you can write with the pen and the Samsung keyboard turns it into text.
15. **iPad / iPhone:** the app works in Safari. Apple Pencil works like a pen: it draws and selects while your finger pans and zooms. Handwriting works with pressure, and Scribble turns handwriting into text in fields.
    - **Install:** Safari → Share → «Add to Home Screen». Do install it: that way the iPad does not delete saved projects and videos if you don't open the app for a while.
    - **Laser:** the iPad doesn't allow Bluetooth connections from web pages (in any browser). Put the meter in keyboard mode, pair it in Settings → Bluetooth and choose «Any meter in keyboard mode».
    - Apple Pencil has no button that a web page can see; for Esc use the app's Esc button.
    - Multiple screens (Screens) are not available on iPad.
16. 📝 **Notes by typing or by voice:** in survey press 📝 Note. Type, or press 🎤 Dictate and speak: what you say becomes text. Then tap the plan on a wall, a corner or inside a room. Dictation also works in photo and video notes.
17. **Numbered pins:** every photo (orange), video (purple) and note (blue) gets a number in the plan. Tap a pin to view it, edit its text, move it or delete it.

18. 📄 **Survey report PDF:** the 📄 PDF button in the survey bar. Fill in project/client, address and who surveyed. The PDF has a summary and rooms with areas, an overview plan with the numbered pins, an index (number, type, position, note, time), the photos, a frame from each video and the notes. Videos can't go inside a PDF; ⬇ Videos downloads them with the same number in the file name.
19. **Ceiling height and wall thickness:** they are on the outline bar and the survey bar (Ceiling height m, Ext. wall cm, Int. wall cm) and change the plan at once.
20. 🏗 **Balconies, awnings, pergolas, stairs:** the Extras button on the plan toolbar, or 🏗 Balcony / stair on the survey bar.
    - **Balcony:** click the exterior wall at the centre of the balcony. Set length, depth (cantilever), offset from the corner and slab thickness. The app asks what railing you want: solid parapet, metal railing, glass, stainless with cables, wooden or none, and the railing height.
    - **Awning:** click the wall above the door or window. Set width, projection, mounting height and fabric (colour or stripes).
    - **Pergola:** click the wall for an attached pergola or a free spot for a free-standing one. Choose wooden, aluminium or bioclimatic, and the cover: open, fabric, reed or louvres.
    - **Stair (maisonette):** click the points of the outer stringer from where the stair goes up and press ✓ Done (or Enter). Set stair width, floor-to-floor height, which side the width goes to and the type (standard, secondary, metal). Like the SKALES tool, it lists the solutions risers × rise / going with 2R+G and marks with ✓ those within limits (2R+G 61–63, R 14–20 cm). Turns get winder steps around the well. Choose railing and tread material.
    - **Move and resize:** click a balcony, awning, pergola or stair to select it (orange). Drag it to move it along the wall. The grips (small squares) change length from either side and depth; a pergola also has a round grip to rotate. Dimensions show while dragging. Arrow keys move 1 cm (Shift 10 cm). Delete removes it. ✎ Properties opens all values as numbers.
    - **Free-shape balcony:** Extras → «Free-shape balcony». Click the wall to start, then the balcony corners, then the wall again: it closes by itself along the wall, even around a building corner. Points snap to the outer wall face, so the balcony never enters the wall. The railing goes only on sides that don't touch the building. Drag a corner to move it; drag the middle of a side to add a corner.
    - **Collision check:** a stair must be inside the room and must not cut interior walls or columns; it may touch a wall. If its width falls into a wall, it flips to the other side by itself. A free-standing pergola that overlaps the building is moved out. A balcony, awning or attached pergola that runs into another wing gets less depth or moves along the wall. If you later change walls and something collides, it shows red with «⚠ collision». The stair railing goes only on free sides, never onto a wall.
    - Everything shows in the plan and is rendered in 3D. Double-click an item to open its properties; the list in the same menu shows what exists.
21. **Outline and survey bars:** they sit at the side so they don't hide the drawing. Drag ⠿ to move them anywhere (double-click: back to default). ▾ collapses them to a small label; tap to expand.
22. **Before / After (renovation):** the survey is the existing state. The first time you go from 📐 Survey to ✏ Design, the app asks whether you want a renovation on a copy.
    - **Yes, renovation:** the Survey is kept as it is and the tabs become Survey · Before and Design · After. Switch with one tap, in plan and 3D. Photos, videos and notes are shared.
    - **No:** keep working on the same design (e.g. new build or just furnishing). If you change your mind, € Cost offers «Make it a renovation».
    - In the renovation plan, **demolished** parts show yellow and **new construction** red; removed openings get ×.
    - The renovation cost automatically includes **Demolitions & removals**: masonry demolition per thickness (m²), new opening cut in an existing wall plus lintel, closing an opening, door and window removal, floor removal where the material changes, kitchen and sanitary removal, debris loading and removal.

23. 🧱 **ETICS per wall:** with Select click the band of an exterior wall (its dark thickness). It is selected in orange and 🧱 ETICS appears below. Choose insulation (white EPS, graphite EPS, XPS, stone wool) and thickness 5–15 cm, or «On all exterior walls». The plan shows a yellow hatched band outside the wall, joined properly at corners. In 3D it is added with the openings left clear. The cost gets: ETICS m² (openings subtracted); reveal lining in running metres (2 jambs + head per opening); mesh corner beads in running metres (2 jambs + head per opening, plus full height at every external building corner); drip profile at the head; window sills; base starter track along the bottom. In a renovation only walls that had no ETICS in the existing state are charged.
24. **Wall thickness and height always at hand:** the plan toolbar has Height (m), Ext. (cm) and Int. (cm) fields on the right.
25. **Room label and quick delete:** with Select drag a room's label anywhere inside the room. When something is selected in the plan (label, corner, wall, opening, column, object), a small bar appears at the bottom with ✕ Delete and ↶ Undo, to remove at once anything placed by mistake.
26. **Column: choose first, then place:** press Column and a small bar opens: shape (rectangular, wall pier, L, T, circular), preset sections, width/depth (plus thicknesses for L/T), angle and **insertion point** (◤ ▲ ◥ / ◀ ● ▶ / ◣ ▼ ◢). Then move the mouse: the column follows as a ghost with the insertion point on the cursor, snapping to corners and walls. Click to place it; you can place several in a row. Esc or × to finish. The bar remembers your settings. To edit an existing one, double-click it: a small properties window opens, where the insertion point stays fixed when you change the size.
27. 📋 **Project details:** + New project (or the first survey) opens a window for project, owner, location/address and engineer; the date fills in automatically (editable) and the engineer is remembered. Edit them from the «Projects» window → «📋 Project details». They go automatically into the survey report and the take-off PDF.
28. ▶ **Quick tour:** the first time you open the app, an 8-step tour shows the tabs, survey, tools, 3D, the selection bar, Undo and cost. Replay it from ⚙ Settings → «Quick tour».
29. **Typed length while dragging (computer):** in Outline or Wall, after the first corner, move the mouse in the wall's direction, type the number (e.g. 4.65) and Enter. The length shows next to the cursor. With Ortho it goes exactly horizontal or vertical. Arithmetic works (e.g. 6.10-1.25). Esc cancels.
30. **Deleting an exterior wall:** select the exterior wall and press ✕ Delete (or Del). The outline opens at that point so you can redraw the piece and press «Close». Openings on other walls stay.
31. **Select and delete:** with Select click something in the plan. Its actions appear in one bar at the bottom: ✎ Properties (where available), ✕ Delete, ⧉ Copy (for objects), ↶ Undo. For corners and walls the bar also has nudge arrows with 1/5/10 cm steps. The Del key also deletes. Undo/Redo are always at the top right.
32. **Millimetre-paper grid:** lines every 1 m, 50 cm, 10 cm, 5 cm and 1 cm, depending on zoom. The point you click snaps exactly to the centimetre. The **0,0** origin is shown with X (red) and Y (green) axes and you start from there; the coordinates at the bottom left are measured from it.
33. **Left panel:** on a computer with one screen it collapses to a «☰ Steps» strip at the left edge. Move the mouse to the edge and it opens; move away and it collapses. 📌 pins it open. On a touch screen, tap the strip to open it and tap outside to collapse it.
34. **Drawing the outline (phone and computer):** with Outline, a bar opens at the bottom showing how many corners you have placed and what to do next.
    - Tap the corners in order. Alternatively, type a **side length** in metres and tap an arrow (← ↑ ↓ →): the next corner goes exactly there. If you have not placed a point yet, the start is placed automatically.
    - **Clean wall junctions:** when an interior wall slightly overshoots or falls short of another wall or the outline (up to 35 cm), its end is trimmed or extended exactly to that wall's line. No overhangs, gaps or spikes remain at junctions, in the plan or in 3D.
    - ↶ Back removes the last point.
    - ✓ Close closes the outline (3 corners or more). Tapping the first corner again does the same.
    - ✕ Cancel clears what you were drawing.

    Once the outline is closed, the bar goes away and you move on to the walls.

35. ⚙ **Settings (gear at the top):**
    - Rotation, scroll (zoom), pan and Walk speed.
    - **Smooth camera motion**: the camera follows with easing, without jerks.
    - **Performance**: «Auto» renders a light image while you move, so it stays fast, and full quality as soon as you stop. «Always full» and «Always fast» are also available.
    - **Button & text size** 85–150%.
    - **App colour**.
    - **High contrast** for low vision.
    - **Reduce motion**: no effects or easing.

    Settings are kept on the device.
36. **Auto-hiding tools (optional):** tools are fixed. If you want them to slide to the edges after 5″ of no use, turn it on from the `Screens` menu.
37. **Solid objects:** when you drag an object, walls and other objects are obstacles: it stops at them and slides along them. It passes through doors. Chairs and stools fit under tables and desks. Kitchens and wall units do not pass through perpendicular walls.
38. **Mouse precision:** a plain click selects without moving anything, because a move starts only after you drag a few pixels. An object is also selected when you click its empty space, e.g. between a chair's legs or inside a lamp frame. For centimetre precision, work in the plan with grips, typed dimensions and arrow keys.
39. **Objects in the plan:** everything you place (furniture, kitchen, bathroom, drywall) also shows in the floor plan with its outline, front side and name. Anything hung high shows dashed.
40. **Objects and drywall in the plan, with grips:** in the plan, with `Select`, click an object (furniture, niche, box-out, bench, cabinet…).
    - Drag it to move. Walls and other objects are obstacles.
    - The square grips on its sides change width and depth while the opposite side stays fixed.
    - The round grip rotates it in 15° steps (Shift for free rotation).
    - Its dimensions and clear distances to the walls are shown. Click any of them for an exact value.
    - Double-click opens width, depth, height, height above floor and angle.
    - Arrow keys move 1 cm (Shift 10 cm). Delete removes and Ctrl+Z undoes.
41. **Wall elevation:** with an object that touches a wall selected, press `▤ Elevation` (or `E`). The wall elevation opens to scale, with a 10 cm grid, the openings and every object on that wall: niches, box-outs, cabinets, sockets, switches.
    - Drag for position and height. The grips change width and the top and bottom edges.
    - Height above floor, distance to the ceiling and left and right distances are shown. Click any dimension for an exact value.
    - Arrow keys 1 cm (Shift 10 cm), `Esc` to close.
42. **3D library:** (automatic replacement of objects with real models is off, because matches are not always good; pick the model yourself) `📚 3D library` in **Furniture** or in the object card. Free CC0 models, free for commercial use too, with pictures, categories (seating, tables, storage, beds, lighting, decor, plants, kitchen & bath, devices, outdoor) and search in Greek or English. From the object card the model takes the object's dimensions. From Furniture a new object is placed at its real size. The choice is saved with the project. Kitchens, islands, wardrobes and drywall are built parametrically, so they don't take a model.
43. **Sign-in with a key:** HEXIS Interior opens only with an **access key** (HXI-XXXX-XXXX-XXXX) from the administrator: 48-hour demo, homeowner access or PRO. Sign in with your email and enter the key once; it is bound to your account.
44. **One device at a time:** if you open it on another device, press “Open here”; the previous one locks within a few minutes. Up to 5 device changes per 30 days; for more, contact the administrator.

45. **No key:** on the “Access key” page press “I don't have a key · request a 48-hour trial”, enter your name, profession and (optionally) phone and message. The administrator approves it in admin (“Interior · licences” → “Trial requests” → “Approve”) and sends the key to your email.
46. The **active balance** is always shown at the top right: units and value in €, e.g. “12 images · €7.92”.
47. After each AI use you see what was used, e.g. “Final image · −1 unit · −€0.66”, or “free · €0”.
48. Press the label for **Activity per use**: when, what, from where, units and €. Failed uses are refunded.
49. 1 unit = 1 final image · 4K = 2 units. Draft images and AI layouts count against the pack's free allowance.
50. In the demo the wallet is local, with simulation buttons to see how the balance goes down.

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## 23 Shortcuts

| | |
|---|---|
| `Ctrl` `Z` / `Ctrl` `Y` | Undo / Redo, in plan and 3D |
| `Enter` | Close the outline |
| `Shift` in plan | Temporary orthogonal drawing |
| `Delete` | Delete the selected furniture, material area, or (in the plan) opening, column, beam, wall |
| `Alt` + drag | Move freely without collisions · in the plan, openings in 1 mm steps |
| Wheel | Zoom in plan and Orbit · forward/back in Walk |
| Right click / `Shift` + drag | Pan in Orbit |
| `R` | Rotate the selected item by 90° |
| `Esc` | Cancel the current drawing or selection |
| Fit | Whole plan on screen |